

Sutton and Barton Exercises Chapter 13

danieltocco0

April 2026

1 Chapter 13

Exercise 13.1. Use your knowledge of the gridworld and its dynamics to determine an *exact* symbolic expression for the optimal probability of selecting the **right** action in Example 13.1.

Solution.

Let $p \in [0, 1]$ denote the probability of selecting **right** in all three nonterminal states (since all states appear identical under the feature representation). Let $\gamma = 1$ and reward $r = -1$ per step.

State layout: S_1 (start) $\rightarrow S_2 \rightarrow S_3 \rightarrow G$ (terminal).

- In S_1 : **right** moves right, **left** causes no movement.
- In S_2 : actions are *reversed* — **right** moves left, **left** moves right.
- In S_3 : **right** moves right (to G), **left** moves left (to S_2).

Expected steps to goal.

From S_3 : with probability p we reach G , with probability $1 - p$ we go to S_2 .

From S_2 : with probability p we go to S_1 , with probability $1 - p$ we go to S_3 .

From S_1 : with probability p we move to S_2 , with probability $1 - p$ we stay in S_1 .

Let v_i denote the (negative) value, i.e., expected number of steps to G , from state S_i . Since reward is -1 per step, $J(\theta) = v_{\pi_\theta}(S_1) = -v_1$.

Setting up the system: since **left** from S_1 causes no movement, the expected steps from S_1 satisfy:

$$v_1 = 1 + (1 - p) v_1 + p v_2 \tag{1}$$

$$v_2 = 1 + p v_1 + (1 - p) v_3 \tag{2}$$

$$v_3 = 1 + (1 - p) v_2 \quad (3)$$

From equation (1):

$$p v_1 = 1 + p v_2 \implies v_1 = \frac{1}{p} + v_2.$$

Substituting into (2):

$$v_2 = 1 + p \left(\frac{1}{p} + v_2 \right) + (1 - p) v_3 = 2 + p v_2 + (1 - p) v_3,$$

$$(1 - p) v_2 = 2 + (1 - p) v_3 \implies v_2 = \frac{2}{1 - p} + v_3.$$

Substituting v_3 from (3):

$$v_3 = 1 + (1 - p) \left(\frac{2}{1 - p} + v_3 \right) = 3 + (1 - p) v_3,$$

$$p v_3 = 3 \implies v_3 = \frac{3}{p}.$$

Back-substituting:

$$v_2 = \frac{2}{1 - p} + \frac{3}{p}, \quad v_1 = \frac{1}{p} + \frac{2}{1 - p} + \frac{3}{p} = \frac{4}{p} + \frac{2}{1 - p}.$$

The performance objective is:

$$J(\theta) = -v_1 = -\frac{4}{p} - \frac{2}{1 - p}.$$

Optimizing over p . Differentiating and setting $\frac{dJ}{dp} = 0$:

$$\frac{dJ}{dp} = \frac{4}{p^2} - \frac{2}{(1 - p)^2} = 0,$$

$$\frac{4}{p^2} = \frac{2}{(1 - p)^2} \implies \frac{(1 - p)^2}{p^2} = \frac{1}{2},$$

$$\frac{1 - p}{p} = \frac{1}{\sqrt{2}} \implies 1 - p = \frac{p}{\sqrt{2}},$$

$$1 = p \left(1 + \frac{1}{\sqrt{2}} \right) = p \cdot \frac{\sqrt{2} + 1}{\sqrt{2}}.$$

$$p^* = \frac{\sqrt{2}}{\sqrt{2} + 1} = \sqrt{2}(\sqrt{2} - 1) = 2 - \sqrt{2} \approx 0.586.$$

This matches the optimal value shown in Example 13.1 (approximately 0.59), with $J(p^*) \approx -11.6$.

Exercise 13.2. Generalize the box on page 199 (the on-policy distribution in episodic tasks), the policy gradient theorem (13.5), the proof of the policy gradient theorem (page 325), and the steps leading to the REINFORCE update equation (13.8), so that (13.8) ends up with a factor of γ^t and thus aligns with the general algorithm given in the pseudocode.

Solution.

Step 1: On-policy distribution with discounting.

Without discounting, the expected number of visits to state s in a single episode satisfies (Equation 9.2):

$$\eta(s) = h(s) + \sum_{\bar{s}} \eta(\bar{s}) \sum_a \pi(a|\bar{s}) p(s|\bar{s}, a),$$

where $h(s)$ is the probability that an episode begins in s . With discounting ($\gamma < 1$), future visitations are worth less, so we include a factor of γ in the second term:

$$\eta(s) = h(s) + \gamma \sum_{\bar{s}} \eta(\bar{s}) \sum_a \pi(a|\bar{s}) p(s|\bar{s}, a).$$

The discounted on-policy distribution is then:

$$\mu(s) = \frac{\eta(s)}{\sum_{s'} \eta(s')}, \quad \text{for all } s \in \mathcal{S},$$

where now $\eta(s)$ encodes discounted visitation counts, so states visited later in the episode receive geometrically smaller weight.

Step 2: Policy gradient theorem with discounting.

The unrolling in the proof of the policy gradient theorem produces transition probabilities $\Pr(s \rightarrow x, k, \pi)$ — the probability of reaching x from s in exactly k steps under π . With discounting, each k -step contribution is weighted by γ^k :

$$\nabla v_\pi(s) = \sum_{x \in \mathcal{S}} \sum_{k=0}^{\infty} \gamma^k \Pr(s \rightarrow x, k, \pi) \sum_a \nabla \pi(a|x, \theta) q_\pi(x, a).$$

Since $\nabla J(\theta) = \nabla v_\pi(s_0)$, and unrolling from s_0 means k -step transitions correspond to timestep $t = k$, we obtain the discounted policy gradient theorem:

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a q_\pi(s, a) \nabla \pi(a|s, \theta), \quad (13.5')$$

where $\mu(s)$ is now the *discounted* on-policy distribution defined in Step 1. The structure of the theorem is unchanged; only the meaning of $\mu(s)$ has been updated to incorporate γ^t weighting.

Step 3: Proof of the discounted policy gradient theorem.

Following the same steps as the undiscounted proof, but carrying γ^k through the unrolling:

$$\begin{aligned}
\nabla J(\boldsymbol{\theta}) &= \nabla v_\pi(s_0) \\
&= \sum_s \left(\sum_{k=0}^{\infty} \gamma^k \Pr(s_0 \rightarrow s, k, \pi) \right) \sum_a \nabla \pi(a|s, \boldsymbol{\theta}) q_\pi(s, a) \\
&= \sum_s \eta(s) \sum_a \nabla \pi(a|s, \boldsymbol{\theta}) q_\pi(s, a) \\
&= \sum_{s'} \eta(s') \sum_s \frac{\eta(s)}{\sum_{s'} \eta(s')} \sum_a \nabla \pi(a|s, \boldsymbol{\theta}) q_\pi(s, a) \\
&= \sum_{s'} \eta(s') \sum_s \mu(s) \sum_a \nabla \pi(a|s, \boldsymbol{\theta}) q_\pi(s, a) \\
&\propto \sum_s \mu(s) \sum_a \nabla \pi(a|s, \boldsymbol{\theta}) q_\pi(s, a). \quad (\text{Q.E.D.})
\end{aligned}$$

The key difference from the undiscounted case is that $\eta(s) = \sum_{k=0}^{\infty} \gamma^k \Pr(s_0 \rightarrow s, k, \pi)$ now discounts visits by when they occur in the episode.

Step 4: REINFORCE update with γ^t .

Starting from (13.5') and writing the gradient as an expectation:

$$\nabla J(\boldsymbol{\theta}) \propto \mathbb{E}_\pi \left[\sum_a q_\pi(S_t, a) \nabla \pi(a|S_t, \boldsymbol{\theta}) \right].$$

Introducing the importance sampling ratio as before:

$$= \mathbb{E}_\pi \left[\sum_a \pi(a|S_t, \boldsymbol{\theta}) q_\pi(S_t, a) \frac{\nabla \pi(a|S_t, \boldsymbol{\theta})}{\pi(a|S_t, \boldsymbol{\theta})} \right].$$

Replacing the sum over a by a sample $A_t \sim \pi$:

$$= \mathbb{E}_\pi \left[q_\pi(S_t, A_t) \frac{\nabla \pi(A_t|S_t, \boldsymbol{\theta})}{\pi(A_t|S_t, \boldsymbol{\theta})} \right].$$

Replacing $q_\pi(S_t, A_t)$ by the sample return G_t (since $\mathbb{E}_\pi[G_t|S_t, A_t] = q_\pi(S_t, A_t)$), and incorporating the γ^t factor from the discounted $\eta(s)$ — which weights timestep t by γ^t relative to $t = 0$ — we obtain:

$$\nabla J(\boldsymbol{\theta}) \propto \mathbb{E}_\pi \left[\gamma^t G_t \frac{\nabla \pi(A_t|S_t, \boldsymbol{\theta})}{\pi(A_t|S_t, \boldsymbol{\theta})} \right].$$

This yields the discounted REINFORCE update:

$$\boxed{\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha \gamma^t G_t \frac{\nabla \pi(A_t|S_t, \boldsymbol{\theta}_t)}{\pi(A_t|S_t, \boldsymbol{\theta}_t)}}. \quad (13.8')$$

The γ^t factor ensures that parameter updates earlier in the episode receive greater weight, consistent with the discounted performance objective $J(\boldsymbol{\theta}) = v_\pi(s_0)$ under $\gamma < 1$. As $\gamma \rightarrow 1$, the factor $\gamma^t \rightarrow 1$ for all t and we recover the undiscounted update (13.8).

Exercise 13.3. In Section 13.1 we considered policy parameterizations using the soft-max in action preferences (13.2) with linear action preferences (13.3). For this parameterization, prove that the eligibility vector is

$$\nabla \ln \pi(a|s, \boldsymbol{\theta}) = \mathbf{x}(s, a) - \sum_b \pi(b|s, \boldsymbol{\theta}) \mathbf{x}(s, b), \quad (13.9)$$

using the definitions and elementary calculus.

Solution.

The softmax policy with linear action preferences is:

$$\pi(a|s, \boldsymbol{\theta}) = \frac{e^{h(s,a,\boldsymbol{\theta})}}{\sum_b e^{h(s,b,\boldsymbol{\theta})}}, \quad h(s, a, \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}(s, a).$$

Rather than applying the quotient rule directly, we first take the logarithm, which separates the numerator and denominator:

$$\ln \pi(a|s, \boldsymbol{\theta}) = h(s, a, \boldsymbol{\theta}) - \ln \sum_b e^{h(s,b,\boldsymbol{\theta})} = \boldsymbol{\theta}^\top \mathbf{x}(s, a) - \ln \sum_b e^{\boldsymbol{\theta}^\top \mathbf{x}(s,b)}.$$

Now we differentiate with respect to $\boldsymbol{\theta}$. The first term is immediate:

$$\nabla [\boldsymbol{\theta}^\top \mathbf{x}(s, a)] = \mathbf{x}(s, a).$$

For the second term, we apply the chain rule. The derivative of $\ln f(\boldsymbol{\theta})$ is $\nabla f(\boldsymbol{\theta})/f(\boldsymbol{\theta})$, so:

$$\nabla \ln \sum_b e^{\boldsymbol{\theta}^\top \mathbf{x}(s,b)} = \frac{\nabla \sum_b e^{\boldsymbol{\theta}^\top \mathbf{x}(s,b)}}{\sum_b e^{\boldsymbol{\theta}^\top \mathbf{x}(s,b)}} = \frac{\sum_b e^{\boldsymbol{\theta}^\top \mathbf{x}(s,b)} \mathbf{x}(s, b)}{\sum_b e^{\boldsymbol{\theta}^\top \mathbf{x}(s,b)}}.$$

Recognizing the softmax in the final expression:

$$= \sum_b \frac{e^{\boldsymbol{\theta}^\top \mathbf{x}(s,b)}}{\sum_{b'} e^{\boldsymbol{\theta}^\top \mathbf{x}(s,b')}} \mathbf{x}(s, b) = \sum_b \pi(b|s, \boldsymbol{\theta}) \mathbf{x}(s, b).$$

Combining both terms:

$$\nabla \ln \pi(a|s, \boldsymbol{\theta}) = \mathbf{x}(s, a) - \sum_b \pi(b|s, \boldsymbol{\theta}) \mathbf{x}(s, b).$$

The second term is the expected feature vector under the current policy $\pi(\cdot|s, \boldsymbol{\theta})$. The eligibility vector therefore measures how much the feature of the

chosen action a deviates from the policy-weighted average feature across all actions — actions whose features differ most from the average receive the largest gradient signal.

Exercise 13.4 — Gaussian Policy Eligibility Vectors Show that for the Gaussian policy parameterization, the eligibility vector $\nabla \ln \pi(a \mid s, \boldsymbol{\theta})$ has the following two parts:

$$\nabla \ln \pi(a \mid s, \boldsymbol{\theta}_\mu) = \frac{1}{\sigma(s, \boldsymbol{\theta})^2} (a - \mu(s, \boldsymbol{\theta})) \mathbf{x}_\mu(s)$$

$$\nabla \ln \pi(a \mid s, \boldsymbol{\theta}_\sigma) = \left(\frac{(a - \mu(s, \boldsymbol{\theta}))^2}{\sigma(s, \boldsymbol{\theta})^2} - 1 \right) \mathbf{x}_\sigma(s)$$

where $\mu(s, \boldsymbol{\theta}) = \boldsymbol{\theta}_\mu^\top \mathbf{x}_\mu(s)$ and $\sigma(s, \boldsymbol{\theta}) = e^{\boldsymbol{\theta}_\sigma^\top \mathbf{x}_\sigma(s)}$.

Solution

Taking the log of the Gaussian policy:

$$\ln \pi(a \mid s, \boldsymbol{\theta}) = -\ln \sigma - \ln \sqrt{2\pi} - \frac{(a - \mu)^2}{2\sigma^2}$$

Part 1: Gradient with respect to $\boldsymbol{\theta}_\mu$

$$\frac{\partial \ln \pi}{\partial \mu} = \frac{a - \mu}{\sigma^2}$$

Since $\mu = \boldsymbol{\theta}_\mu^\top \mathbf{x}_\mu(s)$, we have $\nabla_{\boldsymbol{\theta}_\mu} \mu = \mathbf{x}_\mu(s)$, so by the chain rule:

$$\nabla_{\boldsymbol{\theta}_\mu} \ln \pi = \frac{a - \mu(s, \boldsymbol{\theta})}{\sigma(s, \boldsymbol{\theta})^2} \mathbf{x}_\mu(s)$$

Part 2: Gradient with respect to $\boldsymbol{\theta}_\sigma$

$$\frac{\partial \ln \pi}{\partial \sigma} = -\frac{1}{\sigma} + \frac{(a - \mu)^2}{\sigma^3} = \frac{1}{\sigma} \left(\frac{(a - \mu)^2}{\sigma^2} - 1 \right)$$

Since $\sigma = e^{\boldsymbol{\theta}_\sigma^\top \mathbf{x}_\sigma(s)}$, we have $\nabla_{\boldsymbol{\theta}_\sigma} \sigma = \sigma \mathbf{x}_\sigma(s)$, so by the chain rule:

$$\nabla_{\boldsymbol{\theta}_\sigma} \ln \pi = \frac{1}{\sigma} \left(\frac{(a - \mu)^2}{\sigma^2} - 1 \right) \cdot \sigma \mathbf{x}_\sigma(s) = \left(\frac{(a - \mu(s, \boldsymbol{\theta}))^2}{\sigma(s, \boldsymbol{\theta})^2} - 1 \right) \mathbf{x}_\sigma(s)$$

Exercise 13.5 — Bernoulli-Logistic Unit

A *Bernoulli-logistic unit* is a stochastic neuron-like unit with feature vector $\mathbf{x}(s)$, binary action $A_t \in \{0, 1\}$, and action preferences given by $h(s, 1, \boldsymbol{\theta}) - h(s, 0, \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}(s)$.

Part (a): Show $P_t = \pi(1 \mid S_t, \boldsymbol{\theta}_t) = \sigma(\boldsymbol{\theta}^\top \mathbf{x}(s))$

Applying the softmax distribution (Eq. 13.2) to action $a = 1$:

$$\pi(1 \mid s, \boldsymbol{\theta}) = \frac{e^{h(s,1,\boldsymbol{\theta})}}{e^{h(s,0,\boldsymbol{\theta})} + e^{h(s,1,\boldsymbol{\theta})}}$$

Dividing numerator and denominator by $e^{h(s,1,\boldsymbol{\theta})}$:

$$= \frac{1}{e^{h(s,0,\boldsymbol{\theta})-h(s,1,\boldsymbol{\theta})} + 1} = \frac{1}{1 + e^{-\boldsymbol{\theta}^\top \mathbf{x}(s)}} = \sigma(\boldsymbol{\theta}^\top \mathbf{x}(s)) \quad \square$$

Part (b): Monte-Carlo REINFORCE Update

The general REINFORCE update is:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha G_t \nabla \ln \pi(A_t \mid S_t, \boldsymbol{\theta}_t)$$

Using the eligibility derived in part (c):

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha G_t (A_t - P_t) \mathbf{x}(S_t)$$

Part (c): Eligibility $\nabla \ln \pi(a \mid s, \boldsymbol{\theta})$

Let $P = \pi(1 \mid s, \boldsymbol{\theta}) = \sigma(\boldsymbol{\theta}^\top \mathbf{x}(s))$, so $\pi(0 \mid s, \boldsymbol{\theta}) = 1 - P$.

Case $a = 1$:

$$\nabla \ln \pi(1 \mid s, \boldsymbol{\theta}) = \frac{\nabla P}{P}$$

Using the logistic derivative $\nabla P = P(1 - P) \mathbf{x}(s)$:

$$= \frac{P(1 - P) \mathbf{x}(s)}{P} = (1 - P) \mathbf{x}(s)$$

Case $a = 0$:

$$\nabla \ln \pi(0 \mid s, \boldsymbol{\theta}) = \frac{-\nabla P}{1 - P} = \frac{-P(1 - P) \mathbf{x}(s)}{1 - P} = -P \mathbf{x}(s)$$

Both cases unify as $(a - P) \mathbf{x}(s)$, verified by substitution:

$$a = 1 : (1 - P) \mathbf{x}(s) \checkmark$$

$$a = 0 : (0 - P) \mathbf{x}(s) = -P \mathbf{x}(s) \checkmark$$

Therefore:

$$\nabla \ln \pi(a \mid s, \boldsymbol{\theta}) = (a - \pi(1 \mid s, \boldsymbol{\theta})) \mathbf{x}(s)$$

$\square$